

PAPER_721: UQFF Quantum Variable Documents: r_j, d_g, F_U, f_feedback, Omega_g

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Abstract

Five quantum variable document pages from the UQFF Knowledge Base define the foundational scalar quantities used across all UQFF master equations: r_j (magnetic string distance), d_g (Galactic Center distance), F_U (unified field strength total), $f_{feedback}$ (AGN feedback factor), and Ω_g (Galactic spin rate). Together they define the U_m and U_{bi} coupling to the Milky Way SMBH gravitational system.

1. Variable Registry

Variable	Value	Equation Role
r_j	$1.496 \times 10^{13} \text{ m (1 AU)}$	$U_m = \sum_j \frac{\mu_j}{r_j}(\dots)$
d_g	$2.55 \times 10^{20} \text{ m}$ ($\sim 8 \text{ kpc}$)	$U_{bi} = -\beta_i U_{gi} \Omega_g \frac{M_{bh}}{d_g} \dots$
F_U	$2.28 \times 10^{65} \text{ N}$	$F_U =$ $\sum [k_i U_{gi} - \beta_i U_{gi} \Omega_g \frac{M_{bh}}{d_g} E_{react}] + \dots$
$f_{feedback}$	0.1	$U_{g4} =$ $k_4 \frac{\rho_{SCM} M_{bh}}{d_g} e^{-\alpha t} \cos(\pi t_n) (1 + f_{feedback})$
Ω_g	$7.3 \times 10^{-16} \text{ rad/s}$	$U_{bi} = -\beta_i U_{gi} \Omega_g \frac{M_{bh}}{d_g} \dots$

2. Magnetic String U_m

$$U_m = \sum_j \left[\frac{\mu_j}{r_j} (1 - e^{-\gamma t \cos(\pi t_n)}) \hat{\phi}_j \right] \cdot P_{SCM} \cdot E_{react} \cdot (1 + 10^{13} f_H) \cdot (1 + f_{quasi})$$

3. Buoyancy-Gravity Coupling

$$U_{bi} = -\beta_i U_{gi} \Omega_g \frac{M_{bh}}{d_g} (1 + \varepsilon_{sw} \rho_{sw}) U_{UA} \cos(\pi t_n)$$

References

- UQFF KB grok_share_ba508f76c8e.txt entry #70
- Session 176, v5.33

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